

Shreyas Pathak

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Electrical engineering graduate with a focus on front-end VLSI design, RTL development, verification, and testability. Skilled in SystemVerilog, FPGA prototyping, simulation workflows, and digital logic design with hands-on project experience and ongoing self-learning. Committed to collaboration and delivering precision, reliability, and performance in every design.

Skills

Software & Programming tools: AutoCAD/Solidworks, Matlab/Simulink/ RStudio, UVM, Python/C++/Java, Git, SystemVerilog/Verilog-A/AMS, Gradle, Google Tests, Cadence Virtuoso, LTSpice, Synopsys VCS, Yosys, OpenROAD, Tcl

Hardware & System Design: FPGA programming, Sky130, Raspberry Pi, PLC (LAD, SFC) and HMI programming

Professional & Analytical: Project Management, Risk Management, Data Analysis, Technical Communication

Education

University of British Columbia, Okanagan

September 2020 - April 2025

Bachelor of Applied Sciences - Electrical Engineering (EIT - Engineer In Training)

Relevant Courses: Engineering Drawings and CAD/CAM, Embedded System Design, Energy System Transition, Microelectronics, Motor Drive Systems, Power Electronics, Engineering Project Management

Technical Projects

Full Adder Physical Design, Mentored Internship

July 2025 - September 2025

- Implemented a complete RTL-to-Physical Design flow for a 1-bit full adder, including Verilog coding, Yosys synthesis, and tech mapping with Sky130 standard cells (maj3, xor3).
- Created floorplan, rows, and IO pin placements in OpenROAD, resolving routing track alignment issues by defining custom track grids and TCL-based automation.
- Developed repeatable PnR scripts for placement and routing, integrated DRC/LVS methodology, and completed the RTL to physical design flow with verification, synthesis, and layout generation using the Sky130 PDK.

16-bit Arithmetic Logic Unit (ALU) Design, University of British Columbia, Okanagan

January 2025 - April 2025

- Designed a 16-bit ALU using SystemVerilog, supporting arithmetic/logical operations and a custom 16-bit instruction format with opcode and register addressing.
- Developed a testbench for simulation and verification, analyzing signal behavior with EPWave on EDAPlayground.
- Applied digital design principles and modular HDL practices to implement and verify ALU functionality in a hardware simulation environment.

Analog Integrated Circuit Design Project, University of British Columbia, Okanagan

January 2024 - April 2024

- Developed and optimized low-pass filters and differential amplifiers using Cadence simulation software, achieving precise cutoff frequencies, improved gain stability, and robust circuit designs.
- Conducted detailed parametric analyses to fine-tune resistance, capacitance, and gain, ensuring accurate operation and enhanced reliability under varying circuit conditions.
- Integrated switched-capacitor filters to replace resistors, achieving an estimated 20% improvement in noise reduction and gain stability, particularly in high-frequency applications.

Digital Logic and FPGA Design, University of British Columbia, Okanagan

January 2022 - April 2022

- Designed and simulated combinational and sequential circuits including ripple-carry adders, 2's complement logic, and FSMs using SystemVerilog in Intel Quartus.
 - Developed modular Verilog designs for arithmetic and control logic, validated through simulation and FPGA testing.
 - Optimized circuit designs for accurate signal behavior from logic gates to synthesized hardware.
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Certificates

BCG GenAI

May 2025

- Completed a job simulation for BCG's GenAI team, developing a Python-based chatbot that provided financial insights using rule-based logic and SEC filings (10-K, 10-Q).
 - Applied pandas for data extraction and manipulation, and translated complex financial data into clear, user-friendly responses through structured logic.
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Volunteer Experience

- The Pink Back Project UBCO** - Supported donation drives, care packages, and campus events to reduce inequalities.
- Teach One campaign** - Taught underprivileged individuals, crafted personalized learning plans to empower their future.